

CENTRAL BOARD OF SECONDARY EDUCATION

SAMPLE QUESTION PAPER

Academic Session: 2026–27

Class: 10

Subject: Mathematics

Maximum Marks: 30

Time Allowed: 1 hour

GENERAL INSTRUCTIONS

- **This Question Paper has 5 sections A, B, C, D, and E.**

- Section A has 6 Multiple Choice Questions (MCQs) of 1 mark each.
- Section B has 2 Very Short Answer (VSA) questions of 2 marks each.
- Section C has 2 Short Answer (SA) questions of 3 marks each.
- Section D has 1 Long Answer (LA) question of 5 marks.
- Section E has 1 Case Study Based Question of 4 marks.
- All questions are compulsory. However, an internal choice is provided in some sections.
- Draw neat figures wherever required. Use of calculators is NOT allowed.

SECTION A

Multiple Choice Questions

Q1. The product of a non-zero rational and an irrational number is:

- (A) always irrational
- (B) always rational
- (C) rational or irrational
- (D) one

(1 Mark)

Q2. The pair of equations $y = 0$ and $y = -7$ has:

- (A) one solution
- (B) two solutions
- (C) infinitely many solutions
- (D) no solution

(1 Mark)

Q3. If the HCF of 65 and 117 is expressible in the form $65m - 117$, then the value of m is:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

(1 Mark)

Q4. In $\triangle ABC$, $DE \parallel BC$. If $AD = x$, $DB = x - 2$, $AE = x + 2$ and $EC = x - 1$, then the value of x is:

(A) 1

(B) 2

(C) 3

(D) 4

(1 Mark)

Q5. The distance of the point $P(2, 3)$ from the x -axis is:

(A) 2

(B) 3

(C) 1

(D) 5

(1 Mark)

Q6. If $\sin A = \frac{3}{4}$, then $\tan A$ is:

(A) $\frac{4}{7}$

(B) $\frac{7}{3}$

(C) $\frac{3}{7}$

(D) $\frac{4}{3}$

(1 Mark)

SECTION B

Very Short Answer Questions

Q7. Find the zeroes of the quadratic polynomial $x^2 - 2x - 8$.

(2 Marks)

Q8. For what value of k will the following pair of linear equations have infinitely many solutions?

$$kx + 3y - (k - 3) = 0$$

$$12x + ky - k = 0$$

OR

Find the value of c for which the pair of equations $cx - y = 2$ and $6x - 2y = 3$ will have infinitely many solutions.

(2 Marks)

SECTION C

Short Answer Questions

Q9. Prove that $\sqrt{3}$ is an irrational number.

(3 Marks)

Q10. In the given figure, if $\angle A = \angle B$ and $AD = BE$, show that $DE \parallel AB$.

(3 Marks)

SECTION D

Long Answer Questions

Q11. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train.

OR

Two water taps together can fill a tank in $9\frac{3}{8}$ hours. The tap of larger diameter takes 10 hours less than the smaller one to fill the tank separately. Find the time in which each tap can separately fill the tank.

(5 Marks)

SECTION E

Case Study

Q12. A group of students decided to make a project on "Applications of Trigonometry". They decided to use a kite flying activity to demonstrate some concepts. One student is flying a kite at a height of 60 m from the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 60° .

Based on the above information, answer the following questions:

(i) Find the length of the string, assuming that there is no slack in the string.

(ii) What is the horizontal distance of the kite from the student?

(iii) If the height of the kite is increased to 75 m and the length of the string remains the same, what will be the new angle of inclination with the ground? (Give your answer in terms of trigonometric ratio).

(4 Marks)

End of Question Paper